

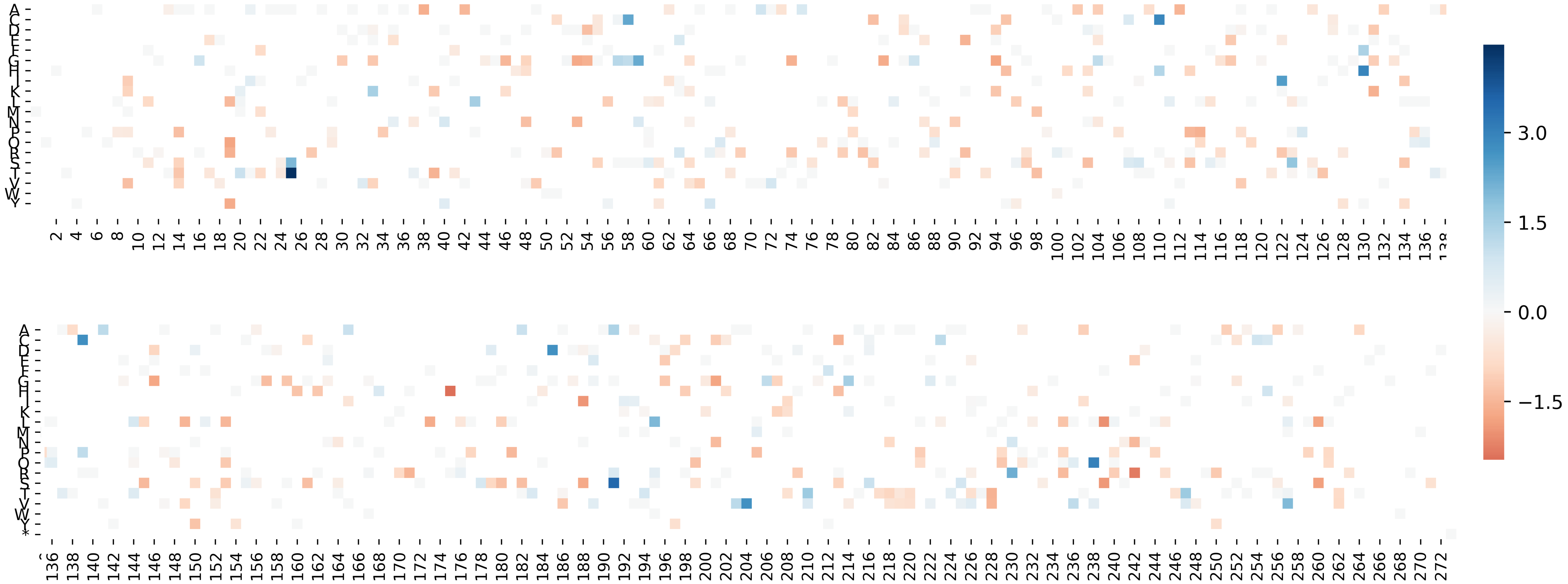
Data set overview

	bin	activity	count	barcode	filename	simple activity	response	num reads	num unique	WT rank	WT percent
0	Bin1	(beta hidroxylation > parent)	20	bc1036--bc1036	bc1036-no-indel.fa.gz	beta	3	168743	13522	0	10
1	Bin2	(beta hidroxylation ~ parent)	54	bc1037--bc1037	bc1037-no-indel.fa.gz	beta	2	159322	12417	0	18
2	Bin3	(beta hidroxylation < parent)	192	bc1038--bc1038	bc1038-no-indel.fa.gz	beta	1	49790	29914	0	2
3	Bin4	(no beta hidroxylation)	694	bc1039--bc1039	bc1039-no-indel.fa.gz	beta	0	77468	58086	1	1
4	Bin5	(gamma hidroxylation > parent)	240	bc1040--bc1040	bc1040-no-indel.fa.gz	gamma	2	107473	60120	1	2
5	Bin6	(gamma hidroxylation ~ parent)	78	bc1030--bc1030	bc1030-no-indel.fa.gz	gamma	1	84390	41407	1	2
6	Bin7	(no gamma hidroxylation)	642	bc1031--bc1031	bc1031-no-indel.fa.gz	gamma	0	105610	68062	0	2
7	Bin8	(azidation > parent)	23	bc1032--bc1032	bc1032--bc1032-no-indel.fa.gz	azid	3	159959	17411	0	10
8	Bin9	(azidation ~ parent)	54	bc1033--bc1033	bc1033--bc1033-no-indel.fa.gz	azid	2	146202	13840	0	17
9	Bin10	(azidation < parent)	177	bc1034--bc1034	bc1034--bc1034-no-indel.fa.gz	azid	1	101769	72886	1	1
10	Bin11	(no azidation)	706	bc1035--bc1035	bc1035--bc1035-no-indel.fa.gz	azid	0	24667	16615	0	2

Slightly concerning WT is the most or second most abundant sequence in all data sets

Ordinal regression on beta-hydroxylation data

Predicted activity of single mutants



Predicted best/worst beta-hydroxylation mutations

Top mutations

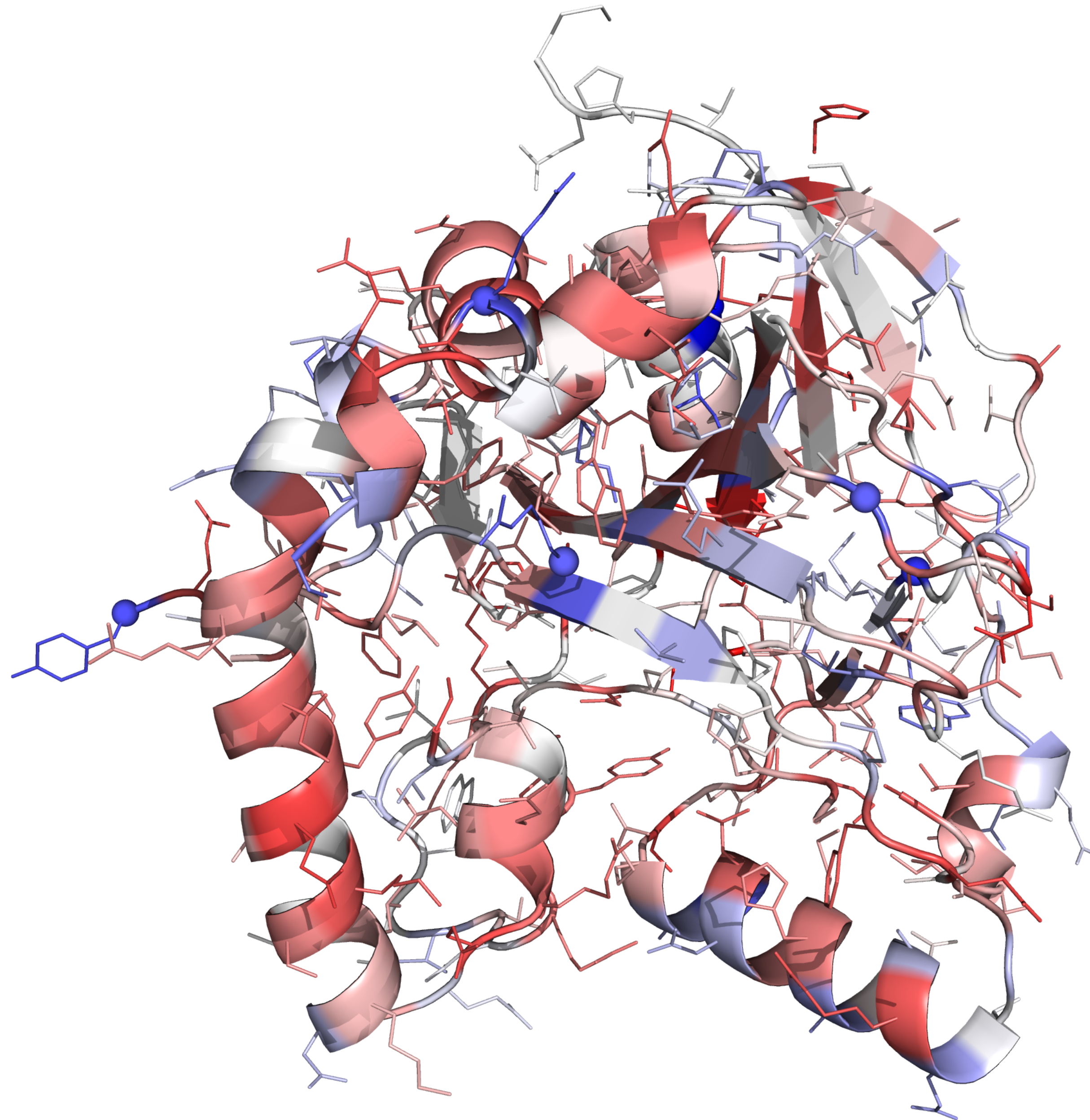
mutation	score
A25T	4.470563
G191S	3.503394
L238Q	3.015973
Y130H	2.932187
R110C	2.915845
R139C	2.711713
A204V	2.706500
G185D	2.701800
K122I	2.474143
S58C	2.307479

Bottom mutations

mutation	score
R175H	-2.472927
K242R	-2.290127
F239L	-2.035796
N188I	-1.981472
F239S	-1.921125
F260S	-1.803252
F260L	-1.787721
H19Q	-1.756778
E94G	-1.755097
S201G	-1.744666

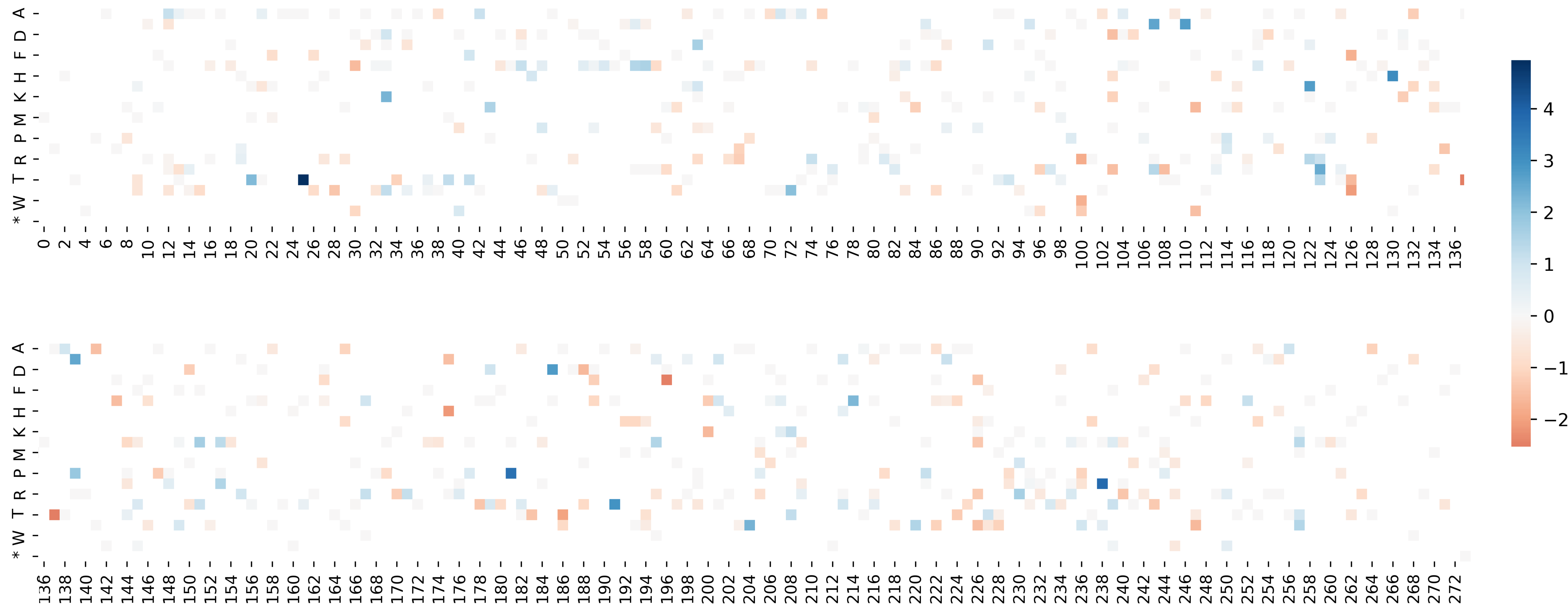
Note: indices may be off by one from standard numbering

Beta-hydroxylation activity mapped onto structure (PDB ID: 3W20)



Ordinal regression on azidation data

Predicted activity of single mutants



Predicted best/worst azidation mutations

Top mutations

mutation	score
A25T	4.936893
L238Q	3.813944
L181P	3.673311
Y130H	3.081721
G191S	2.948203
G185D	2.799994
R110C	2.774949
K122I	2.730002
R107C	2.580657
R139C	2.573946

Bottom mutations

	mutation	score
194	A137T	-2.526992
269	D196E	-2.495752
240	R175H	-2.134964
181	I126V	-2.104788
253	A186T	-1.976399
135	C100R	-1.821916
136	C100W	-1.760573
179	I126F	-1.739426
275	E200K	-1.607603
361	I247V	-1.581184

Note: indices may be off by one from standard numbering

Azidation activity mapped onto structure (PDB ID: 3W20)

